

Highlights

Calibrated Gating for Adaptive RAG: A Conformal Bound on the Decision to Retrieve

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- The retrieve/skip gate in adaptive RAG is recast as risk-controlled selection.
- Conformal risk control bounds the harmful-skip rate at a user-chosen tolerance.
- Fixed-threshold gating overshoots the hallucination budget by 34–58% under shift.
- The conformal skip rate auto-scales with model competence at a fixed tolerance.
- Marginal CRC suffices with a calibrated score; Mondrian only for label shift.

Calibrated Gating for Adaptive RAG: A Conformal Bound on the Decision to Retrieve

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ABSTRACT

Adaptive retrieval-augmented generation (RAG) skips retrieval when a language model is judged confident enough to answer from parametric memory, trading accuracy for cost and latency. The component that makes this choice, the retrieve/skip gate, is safety-critical: a skip taken when the model would hallucinate sends an unsupported answer to the user. Existing adaptive-RAG gates (FLARE, Self-RAG, Adaptive-RAG) tune a confidence threshold offline, provide no formal guarantee on this harmful-skip event, and, as we show, drift across distributions. We reframe the gate as a risk-controlled selective-prediction decision under asymmetric cost and apply Conformal Risk Control (CRC). Given an ignorance score that estimates the probability the model answers wrongly without retrieval, the gate skips the lowest-scoring queries, and the threshold is calibrated so the expected harmful-skip rate stays below a user-chosen tolerance. Because the calibration is conformal, the bound is finite-sample and distribution-free. The decisive move is what we conformalize: the gate decision, not the generated answer. On two model families (Qwen2.5-3B, Phi-3.5-mini) across TriviaQA and PopQA, the realized rate tracks the target dial almost exactly from a 2% to a 20% tolerance. A fixed threshold calibrated on one distribution overshoots the hallucination budget by 34–58% on another, whereas per-distribution conformal calibration restores control. The skip rate auto-scales with model competence (35% versus 18% skips at a fixed tolerance). We characterize when group-conditional (Mondrian) calibration is needed, only under label shift given features and not proportion shift, and validate the gate end-to-end with a live retriever as a guaranteed cost/accuracy dial.

1. Introduction

Retrieval-augmented generation (RAG) grounds large language models (LLMs) in external evidence and is now the dominant approach for knowledge-intensive question answering (Lewis et al., 2020; Gao et al., 2023). Retrieval, however, is not free: it adds latency, monetary cost, and, when irrelevant passages are returned, can even degrade an answer the model would have produced correctly on its own. Adaptive RAG therefore makes retrieval optional, invoking it only when the model appears uncertain (Jiang et al., 2023; Asai et al., 2024; Jeong et al., 2024). The component that decides is a binary retrieve/skip gate.

This gate is a safety-critical decision. When it skips retrieval on a query the model can in fact answer, it saves cost at no risk. But when it skips on a query the model would get wrong, the unsupported and often confident answer reaches the user. We call this event a harmful skip; it is precisely the hallucination that retrieval was meant to prevent. The two error directions are not symmetric: an unnecessary retrieval costs a bounded amount of latency, while a harmful skip costs a hallucination. Treating the gate as an ordinary binary classifier, optimized for accuracy, ignores this asymmetry.

Existing adaptive-RAG gates fix a confidence threshold (on token probability (Jiang et al., 2023), a learned critic token (Asai et al., 2024), or a query-complexity classifier (Jeong et al., 2024)) tuned once on a calibration set. None bounds the harmful-skip rate, and none accounts for the fact that the threshold’s behavior depends on the deployment

distribution. We demonstrate this failure directly: a threshold calibrated on PopQA, applied to TriviaQA, overshoots a 10% hallucination budget by 34% for Qwen2.5-3B and 58% for Phi-3.5-mini (Section 5.2). The model is more competent on TriviaQA, so the score distribution shifts, and the fixed threshold silently becomes unsafe.

Our approach. We reframe the retrieve/skip gate as a risk-controlled selective-prediction decision and bring Conformal Risk Control (CRC) (Angelopoulos et al., 2024) to bear on it. Given an ignorance score that estimates the probability the model answers a query wrongly without retrieval, we skip the queries it scores as most confident. We calibrate the skip threshold on a calibration set so that the expected harmful-skip rate is bounded by a user-chosen tolerance. Because CRC is conformal, the bound is finite-sample and distribution-free: it holds for any score, any model, and any data distribution, provided calibration and deployment data are exchangeable. The single move that makes this work is what we conformalize: not the generated answer, as in prior conformal-RAG and conformal-LLM work (Kumar et al., 2023; Quach et al., 2024), but the gate decision itself (Figure 1). The guarantee is placed on the input to generation (whether to retrieve) rather than its output.

Contributions.

- (1) We formalize the retrieve/skip gate as a risk-controlled selective-prediction decision under asymmetric cost, identifying the gate decision (not the generated answer) as the right object to conformalize, and derive a CRC-calibrated gate that bounds the expected harmful-skip rate at any user-chosen tolerance (Section 3).

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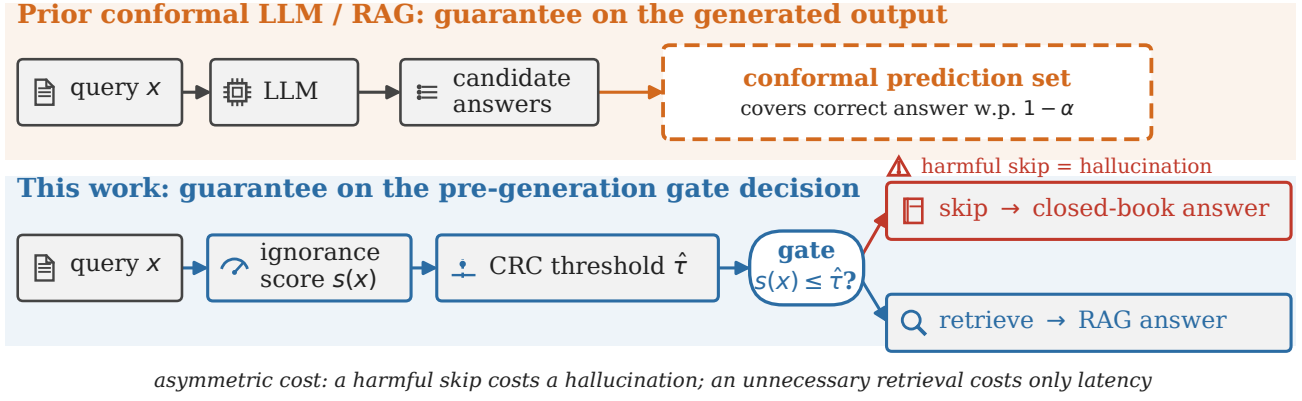


Figure 1: What we conformalize. Prior conformal LLM/RAG methods place the guarantee on the generated output, wrapping the answer in a prediction set that covers the correct answer with high probability (top). We instead place the guarantee on the pre-generation gate decision (bottom): an ignorance score feeds a CRC-calibrated threshold that chooses skip versus retrieve, bounding the harmful-skip rate (a skip on a query the model gets wrong, i.e. a hallucination) at a user-chosen tolerance. The cost is asymmetric: a harmful skip costs a hallucination, whereas an unnecessary retrieval costs only latency.

- (2) We show on real LLM behavior that the realized harmful-skip rate tracks the target dial almost exactly from a 2% to a 20% tolerance, whereas the prevailing fixed-threshold practice overshoots the hallucination budget by 34–58% under distribution shift, which per-distribution conformal calibration corrects (Sections 5.1–5.2).
- (3) We show the conformal skip rate auto-scales with model competence at a fixed tolerance, turning skip rate into an emergent measure of what the model knows, and that every result replicates across two model families (Section 5.3).
- (4) We characterize when group-conditional (Mondrian) calibration is needed, only under label-shift-given-features and not proportion shift, and validate the gate end-to-end with a live retriever as a guaranteed cost/accuracy dial (Sections 5.4–5.5).

2. Related work

2.1. Retrieval-Augmented Generation and adaptive retrieval

Retrieval-Augmented Generation (Lewis et al., 2020) grounds LLM inference in external evidence, substantially improving factual accuracy on knowledge-intensive tasks (Gao et al., 2023; Karpukhin et al., 2020). Unconditional retrieval, however, adds latency and cost, and context that is irrelevant to a query the model already knows can degrade accuracy (Yan et al., 2024). This motivates adaptive retrieval strategies.

FLARE (Jiang et al., 2023) monitors generation-time token probability and retrieves whenever it falls below a hand-tuned threshold. Self-RAG (Asai et al., 2024) trains the LLM itself to emit a special reflection token ([Retrieve] / [No

Retrieve]), framing the gate as a generation task. Adaptive-RAG (Jeong et al., 2024) trains a separate complexity classifier to route each query to no-retrieval, single-step, or multi-step retrieval. DRAGIN (Su et al., 2024) operates at the token level, triggering retrieval when the entropy of the next-token distribution exceeds a threshold. Corrective RAG (Yan et al., 2024) adds a lightweight retrieval evaluator that judges retrieved document quality and, when documents are deemed unreliable, escalates to a web search, introducing a form of retrieval confidence without a formal guarantee.

All of these methods fix or learn a decision boundary on a training or calibration distribution, without providing any guarantee on the harmful-skip error rate, and without recalibrating to the deployment distribution. We demonstrate in Section 5.2 that this failure is measurable and systematic: a threshold calibrated on one dataset overshoots the hallucination budget by 34–58% on another. Our contribution is orthogonal to the choice of uncertainty signal: FLARE’s token probability, Self-RAG’s reflect score, or DRAGIN’s entropy can all serve as features in our ignorance score, after which CRC supplies the missing finite-sample guarantee.

2.2. Conformal prediction

Conformal prediction (Vovk et al., 2005) constructs prediction sets that contain the true label with user-specified probability under exchangeability, with no assumptions on the data distribution. Angelopoulos and Bates (2023) provide a modern tutorial. Conformal Risk Control (CRC) (Angelopoulos et al., 2024) generalizes coverage to any monotone, bounded loss L , bounding $\mathbb{E}[L] \leq \alpha$ via a one-dimensional threshold search, the formulation we use for harmful-skip control. Risk-Controlling Prediction Sets (Bates et al., 2021) provide a training-conditional (PAC-style) variant; the expected-value CRC guarantee we use is weaker but requires no learning. Group-conditional (Mondrian) conformal prediction (Vovk, 2012) delivers per-group validity; covariate-shift corrections (Tibshirani et al., 2019)

handle distribution mismatches. We use these to characterize precisely when marginal CRC is insufficient (Section 5.4).

2.3. Conformal prediction for LLMs and RAG

Conformalizing the answer. The majority of conformal-LLM work controls a property of the model’s output. Kumar et al. (2023) construct prediction sets for multiple-choice QA that cover the correct option with high probability. Quach et al. (2024) develop conformal language modeling, stopping sequential generation once a coverage guarantee is satisfied. TRAQ (Li et al., 2024) is the most directly related prior work: it applies conformal prediction end-to-end to RAG at both retriever and generator stages, guaranteeing with high probability that the prediction set contains a semantically correct answer. Ren et al. (2023) apply conformal prediction to LLM-based robot planning. Xiong et al. (2024) show that eliciting calibrated confidence from an LLM is non-trivial and model-dependent.

Conformalizing abstention. Closest in spirit to our work is Abbasi Yadkori et al. (2024), who apply conformal abstention at the generation level: the model outputs “I don’t know” when its self-consistency score is below a CRC-calibrated threshold, bounding the hallucination rate on non-abstained queries. Our gate operates at the retrieval level, deciding whether to retrieve before any generation occurs, and its asymmetric cost formulation (a harmful skip costs a hallucination; an unnecessary retrieval costs only latency) is specific to the RAG setting.

The critical distinction. All prior conformal-LLM work controls a property of the model’s output or post-generation decision; we control the pre-generation input decision: whether to retrieve. This is a structurally distinct guarantee (applied to the binary gate rather than the answer) that, to our knowledge, has not previously been addressed.

2.4. Uncertainty estimation for LLMs

Informative uncertainty signals are available across multiple modalities. Token log-probability is a strong scalar baseline and emerges as genuine self-knowledge in capable LLMs (Kadavath et al., 2022); calibration of neural confidences under distribution shift is a known challenge (Guo et al., 2017; Xiong et al., 2024). Self-consistency (Wang et al., 2023) aggregates multiple samples to reduce surface-form variability. Semantic entropy (Kuhn et al., 2023) estimates uncertainty at the meaning level, avoiding spurious variance from paraphrase; Farquhar et al. (2024) demonstrate this is effective for hallucination detection in real LLMs. Black-box consistency checking via SelfCheckGPT (Manakul et al., 2023) requires no access to model logits. Ji et al. (2023) survey LLM hallucination broadly. We use a multivariate logistic model over these signals (Section 3.2); a key empirical finding is that signal granularity (density of distinct score values) matters more than discriminative power for CRC efficiency, a design principle absent from prior uncertainty-estimation work.

2.5. Selective prediction

Selective prediction (Geifman and El-Yaniv, 2017) frames a coverage/accuracy trade-off for an abstaining classifier. We apply this lens to the retrieval decision: the model “abstains” from answering without retrieval when uncertain, handing off to the retriever as its safe fallback. Unlike classical selective prediction, our cost structure is asymmetric: the cost of a harmful abstention (skipping retrieval when the model is wrong) far exceeds the cost of unnecessary retrieval (latency). This asymmetry motivates CRC over standard coverage-maximizing selective prediction (Geifman and El-Yaniv, 2017).

3. Method

3.1. The gate as risk-controlled selective prediction

Let x be a query, M an LLM, and R a retrieval pipeline. A gate $d(x) \in \{\text{SKIP}, \text{RETRIEVE}\}$ decides whether M answers x from parametric memory (SKIP) or with retrieved evidence (RETRIEVE). Let $w(x)$ be the indicator that M answers x wrongly without retrieval. The safety-critical event is the harmful skip,

$$h(x) = \mathbf{1}[d(x) = \text{SKIP}] \cdot w(x), \quad (1)$$

a skip taken on a query the model gets wrong, an avoidable hallucination. The cost structure is asymmetric: an unnecessary retrieval ($d = \text{RETRIEVE}$, $w = 0$) costs bounded latency, while a harmful skip costs an ungrounded answer. We therefore do not seek the accuracy-optimal gate; we seek the gate that minimizes retrieval subject to a cap on harmful skips:

$$\min_d \mathbb{E}[\mathbf{1}[d(x) = \text{RETRIEVE}]] \quad \text{s.t.} \quad \mathbb{E}[h(x)] \leq \alpha, \quad (2)$$

for a user-chosen tolerance $\alpha \in (0, 1)$. This is selective prediction (Geifman and El-Yaniv, 2017) applied to retrieval, with the abstention direction (here, RETRIEVE) treated as the safe fallback.

3.2. Ignorance score

We summarize the model’s per-query ignorance with a scalar $s(x) \in [0, 1]$, higher meaning more likely wrong without retrieval. From $k = 8$ closed-book samples we extract four features: negative mean token log-probability of the greedy answer (Kadavath et al., 2022); semantic entropy over normalized-answer clusters (Kuhn et al., 2023); one minus self-consistency (the fraction of samples agreeing with the greedy answer) (Wang et al., 2023); and answer length. A multivariate logistic IGNORANCEMODEL maps these to $\hat{P}(w = 1 \mid x)$, which is $s(x)$.

Two empirical facts shape this design (Sections 5.1–5.2). First, the individual signals have nearly identical discriminative power (AUROC for predicting w ranges 0.85–0.89; Table 2), so the choice among them is not about information content. Second, what matters for CRC is granularity: a discrete score with few distinct values cannot place a threshold that spends the budget precisely. Discrete semantic entropy

from $k = 8$ samples is too coarse, spending only $\approx 3.7\%$ of a 10% budget, whereas the continuous log-probability (and the logistic score that inherits its granularity) uses the budget fully at the same AUROC. We therefore use the continuous multivariate score throughout.

3.3. Conformal Risk Control of the gate

The gate is a threshold rule, $d(x) = \text{SKIP} \iff s(x) \leq \tau$. Smaller τ skips fewer (only the most confident) queries, so the harmful-skip rate is monotone non-decreasing in τ . Given a calibration set $\{(s_i, w_i)\}_{i=1}^n$ of scored, graded queries, we set

$$\hat{\tau} = \max \left\{ \tau : \frac{1 + \sum_{i=1}^n \mathbf{1}[s_i \leq \tau] w_i}{n+1} \leq \alpha \right\}. \quad (3)$$

Under exchangeability of calibration and test data, Conformal Risk Control (Angelopoulos et al., 2024) guarantees

$$\mathbb{E}[h(x_{\text{test}})] \leq \alpha, \quad (4)$$

where the expectation is over the draw of calibration and test points. The $+1$ in the numerator and $n+1$ in the denominator are the finite-sample correction; they make the bound hold exactly at any n , at the cost of slight conservatism that vanishes as $n \rightarrow \infty$. This is a correctness property, preventing the small-sample overshoot that an uncorrected empirical quantile incurs, rather than the efficiency headline. Since the harmful-skip rate is monotone non-decreasing in τ , $\hat{\tau}$ is the largest threshold satisfying the constraint in (2), hence it minimizes retrieval frequency subject to the guarantee.

Exchangeability holds when calibration and deployment queries are drawn i.i.d. from the same distribution, the standard assumption for split-conformal procedures. When a known covariate shift is present, the weighted extension of Tibshirani et al. (2019) restores validity at the cost of a density-ratio estimate over the shift covariates.

3.4. Baselines

We compare the CRC gate against: **never-retrieve** and **always-retrieve** (the trivial endpoints); **tuned-threshold**, the empirical α -quantile of $\{s_i : w_i = 1\}$ with no finite-sample correction; **fixed-threshold**, the same threshold calibrated on one distribution and deployed on another without re-calibration, representing any gate that does not adapt its threshold at deployment time; and **group-conformal (Mondrian)**, one CRC threshold per novelty bucket so that $\mathbb{E}[h \mid \text{bucket}] \leq \alpha$ within each bucket.

3.5. When is group-conditional control needed?

Marginal CRC (3) bounds the overall expected harm. Under distribution shift one might fear it fails on the worst sub-population. We show (Section 5.4) that when the score is approximately conditionally calibrated ($s(x) \approx P(w = 1 \mid x)$, which a multivariate logistic score on informative features achieves), marginal CRC already yields near-per-bucket control, because the score captures the very feature axes along which the distribution shifts. Group-conditioning becomes necessary only under label-shift-given-features:

when two groups share feature values but differ in $P(w \mid \text{features})$, so the score cannot tell them apart. That case requires knowing the shift axis in advance and pre-declaring buckets.

4. Experimental setup

Models and data. We use two instruction-tuned LLMs from different families, Qwen2.5-3B-Instruct (Qwen Team, 2024) and Phi-3.5-mini-instruct (Abdin et al., 2024), in fp16 on a single GPU. We evaluate on two open-domain QA datasets with contrasting difficulty: PopQA (Mallen et al., 2023) (test split, popularity-stratified 1000 questions; long-tail-heavy) and TriviaQA (Joshi et al., 2017) (rc.nocontext validation, 1000 questions; more head entities). Closed-book accuracy is 0.126/0.365 (Qwen) and 0.140/0.434 (Phi) on PopQA/TriviaQA, giving a wide competence gap to test adaptive behavior.

Protocol. For each query we draw $k = 8$ samples (one greedy, seven temperature-sampled), grade correctness by exact match against gold answer aliases, and extract the four ignorance features. We fit the IGNORANCEMODEL on a held-out fold and report all gate metrics over 10 random calibration/test splits ($n_{\text{cal}} \approx 333$, one-third of the 1000-question pool; CRC remains valid at any n by construction, as the finite-sample correction in (3) absorbs small- n variance), as mean $\pm$ std. AUROC values are pooled across datasets.

End-to-end retriever. For the downstream evaluation (Section 5.5) we attach a live retriever (MediaWiki subject-page search; context-hit 100%, answer-in-context $\approx 84\%$ on PopQA). A skip yields the closed-book answer; a retrieve yields the RAG answer over the retrieved context.

Synthetic shift probe. To exhibit the regime where Mondrian is required (Section 5.4) we use a controlled generator with a known $P(w \mid x)$ and two novelty buckets (popular, $p_{\text{know}} = 0.85$; long-tail, $p_{\text{know}} = 0.25$), calibrating on a 20% long-tail mixture and deploying on a 70% long-tail mixture (200 draws).

5. Results

5.1. The safety guarantee holds and is not conservative

Figure 2 and Table 1 report the realized harmful-skip rate as the target α is swept from 0.02 to 0.20. On both model families the realized rate tracks the dial almost exactly: for example, for Qwen2.5-3B (pooled), $\alpha = 0.02 \rightarrow 0.05 \rightarrow 0.10 \rightarrow 0.15 \rightarrow 0.20$ yields realized $0.019 \rightarrow 0.049 \rightarrow 0.098 \rightarrow 0.149 \rightarrow 0.199$, and Phi-3.5-mini matches to within a point at every setting. The guarantee is therefore not merely satisfied but tight: the gate spends the budget rather than hiding behind needless retrieval. Retrieval frequency falls monotonically as α rises (Figure 2b), so α acts as a single interpretable cost dial.

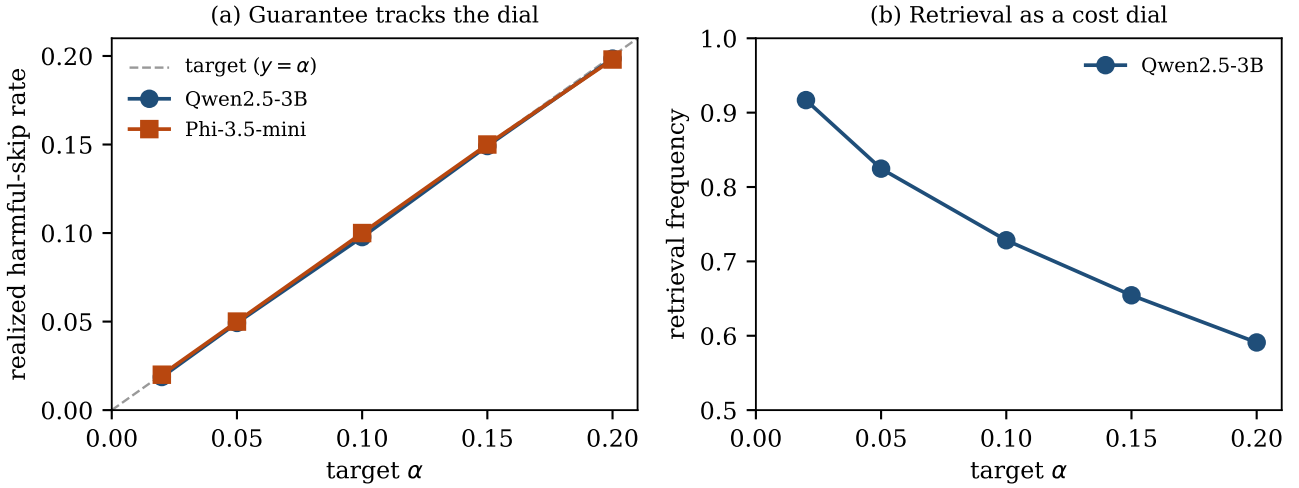


Figure 2: The conformal guarantee on real LLM behavior. (a) Realized harmful-skip rate vs. target α for both model families; the dashed line is $y = \alpha$. (b) Retrieval frequency falls monotonically as the budget loosens, so α is an interpretable cost dial. Mean over 10 calibration/test splits.

Table 1

Coverage dial (Qwen2.5-3B, pooled). Realized harmful-skip rate tracks target α ; retrieval trades off monotonically.

target α	0.02	0.05	0.10	0.15	0.20
realized harm	0.019	0.049	0.098	0.149	0.199
retrieval freq	0.917	0.825	0.728	0.654	0.591

Table 2

AUROC of ignorance signals for predicting wrongness (pooled, TriviaQA+PopQA). Discriminative power is near-identical; the operative difference is score granularity, not information (Section 3.2). Dashes indicate signals not computed for Phi-3.5-mini: the $k = 8$ sampling budget for semantic entropy and self-consistency was allocated to Qwen for the primary granularity analysis; accordingly the IGNORANCEMODEL for Phi-3.5-mini uses neg. log-prob and answer length only, consistent with its AUROC of 0.888.

signal	Qwen2.5-3B	Phi-3.5-mini
neg. mean log-prob	0.881	0.888
semantic entropy	0.861	n/a
1-self-consistency	0.854	n/a

5.2. Fixed-threshold gating is unsafe under distribution shift

The central practical claim is that the prevailing fixed-threshold practice silently breaks under distribution change. Table 3 fixes $\alpha = 0.10$ and compares gates. A threshold calibrated on PopQA and deployed on TriviaQA realizes harm 0.134, a 34% overshoot of the budget in the dangerous direction; the reverse calibration is over-conservative (harm 0.069, wasting retrieval). Per-distribution conformal calibration hits the target on each distribution (0.098 on TriviaQA, 0.097 on PopQA). The same failure is worse on

Phi-3.5-mini: the PopQA $\rightarrow$ TriviaQA threshold overshoots by 58%. The reason is exactly the competence gap: the score distribution shifts between datasets, so a threshold frozen on one is mis-calibrated on the other. Conformal recalibration, being distribution-free, adapts automatically.

5.3. Adaptive efficiency and generality

At a fixed budget $\alpha = 0.10$, the gate skips more on the dataset the model knows better, with no re-tuning (Table 4). Qwen2.5-3B skips 35% of TriviaQA but only 18% of PopQA; Phi-3.5-mini skips 42% vs 20%. The skip rate is thus an emergent measure of model competence: the more the model knows, the more retrieval the gate safely declines, all while holding the same hallucination guarantee. Every core result (the coverage dial, the fixed-threshold failure, and adaptive skipping) replicates across both model families, indicating the findings are not model-specific.

5.4. When is group-conditional control needed?

Figure 3(a) contrasts two shift regimes at $\alpha = 0.10$, reporting harm on the long-tail sub-population. Under a synthetic label-shift (calibration 20% $\rightarrow$ deployment 70% long-tail, with bucket-dependent $P(w | x)$), marginal CRC blows the budget on the long tail (harm 0.243, $2.4 \times \alpha$) while group-conditional CRC restores control (harm 0.094), at modestly higher retrieval. This is the textbook case for Mondrian. On real data, however, the same 20% $\rightarrow$ 70% long-tail proportion shift leaves marginal CRC barely over budget (long-tail harm 0.113 vs group 0.091), and across four real

Table 3

What conformal gating buys ($\alpha = 0.10$). A fixed threshold (calibrated once, deployed without re-calibration to the new domain) drifts with the deployment distribution; only per-distribution conformal calibration hits the target with a guarantee. The fixed-threshold rows model any deployment that does not adapt its threshold at the target domain. Harm = realized harmful-skip rate.

gate	TriviaQA	PopQA	controlled?
never-retrieve	0.635	0.874	no (catastrophic)
always-retrieve	0.000	0.000	safe, retr. = 1
fixed thr. (PopQA $\rightarrow$ Triv.)	0.134	n/a	no, +34%
fixed thr. (Triv. $\rightarrow$ PopQA)	n/a	0.069	no, over-cons.
conformal (per-dist.)	0.098	0.097	yes

Table 4

Adaptive efficiency and cross-family generality ($\alpha = 0.10$). Skip rate scales with closed-book competence at a fixed hallucination guarantee.

dataset	Qwen2.5-3B		Phi-3.5-mini	
	acc	skip	acc	skip
TriviaQA	0.365	35%	0.434	42%
PopQA	0.126	18%	0.140	20%

shift configurations marginal CRC never violates α by more than 0.013. The reason is that the multivariate ignorance score is approximately conditionally calibrated, so proportion shifts in buckets it can already distinguish do not break marginal control. The design guidance is therefore precise: use marginal CRC with a good score; reserve Mondrian for known label-shift-given-features.

5.5. End-to-end validation with a live retriever

Attaching a real retriever (Section 4) lets us measure final answer accuracy against retrieval cost on PopQA (Figure 3(b)). Three reference points frame the result: never-retrieve 0.126, always-retrieve 0.612, and an oracle gate 0.628 at 0.874 retrieval. The ignorance-score gate dominates uninformed (random) skipping at every retrieval budget, confirming the score is end-to-end meaningful, and the oracle’s edge over always-retrieve shows there is real headroom for gating.

We are deliberately careful about what this does and does not show. On PopQA the retrieval fix-rate (0.57, the chance

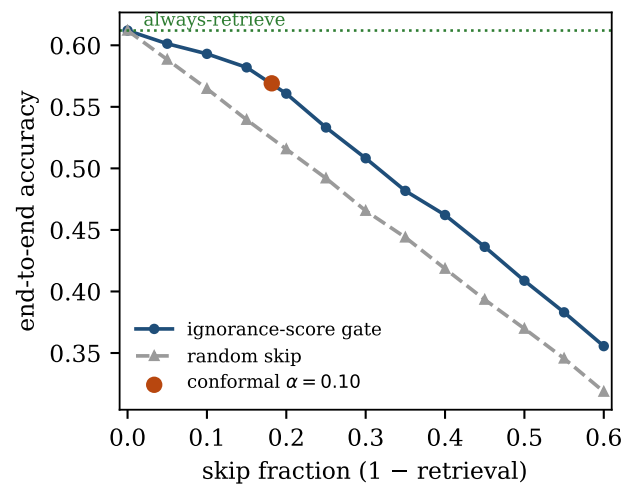
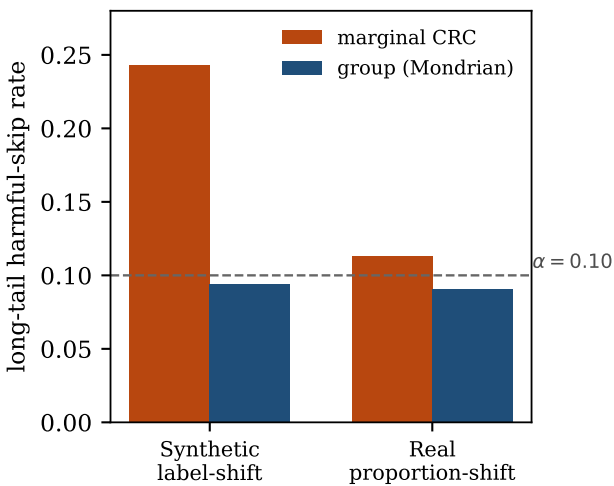


Figure 3: (a) When group-conditioning is needed ($\alpha = 0.10$, long-tail harm). Under synthetic label-shift, marginal CRC violates the budget and Mondrian is required; under real proportion-shift, a conditionally-calibrated score keeps marginal CRC near α without grouping. (b) End-to-end accuracy vs. skip fraction on PopQA with a live retriever. The ignorance-score gate beats random skipping at every budget; the always-retrieve line and the conformal $\alpha = 0.10$ operating point are marked. The gate is a guaranteed cost/accuracy dial.

retrieval rescues a wrong closed-book answer) far exceeds the poison-rate (0.13, the chance it spoils a correct one), and only 12.6% of questions are answerable closed-book. Retrieval is thus beneficial even on the queries the gate would skip, so selective retrieval here cannot beat full RAG on accuracy; the conformal gate is a guaranteed cost/accuracy dial (e.g. cut 18% of retrievals for -4.3 accuracy points, with harmful skips bounded by α), not an accuracy win. The accuracy-positive regime requires a workload with a genuine closed-book-answerable head, where retrieval is wasteful or harmful. Analytically, the gate beats full RAG exactly when the poison avoided on skipped-known queries exceeds the fixes lost on skipped-unknown queries, a condition our PopQA numbers do not meet. We report the downstream result honestly as cost control with demonstrated headroom, not as an accuracy improvement.

6. Discussion

A dial, not just a gate. FLARE and Self-RAG expose one operating point per threshold; the conformal gate exposes a controlled frontier indexed by an interpretable risk level α . An operator who can state an acceptable hallucination-from-skipping rate gets the cheapest gate that respects it, with a finite-sample guarantee and automatic adaptation to the deployment distribution. In practice, α can be read directly from Table 1: setting $\alpha = 0.05$ permits roughly 1 in 20 skipped queries to cause hallucination and retrieves $\approx 82\%$ of the time; relaxing to $\alpha = 0.10$ saves an additional ≈ 10 percentage points of retrieval at the cost of doubling the tolerated harmful-skip rate. Operators with an explicit cost budget can therefore select α from the retrieval-frequency row rather than hand-tuning a confidence threshold.

Granularity over AUROC. A practical lesson from Section 3.2 is that the standard signal-selection metric, AUROC, is insufficient for choosing a gate score. Two signals with equal AUROC can differ sharply in how precisely CRC can spend the budget, because a coarse, few-valued score cannot place a threshold near the target rate. Practitioners should prefer continuous or multivariate scores for risk-controlled gating.

Limitations. (i) CRC controls the expected harmful-skip rate over random draws of the calibration set, not a high-probability per-deployment bound. Concretely, with $n_{\text{cal}} = 333$ and $\alpha = 0.10$, there is non-negligible probability that a given calibration draw produces a threshold whose realized harmful-skip rate exceeds α on that deployment. A training-conditional $(1 - \delta)$ variant (Bates et al., 2021) would supply an explicit δ -failure probability at the cost of a looser threshold; we consider this natural future work for settings where even a small probability of budget violation is unacceptable. (ii) Our semantic entropy uses normalized-string clustering, a discrete proxy for NLI-based semantic entropy (Kuhn et al., 2023); a finer variant may recover budget-spending granularity from that signal. (iii) Wrongness labels use exact match against gold aliases, appropriate for short-form QA;

judge-noise sensitivity on long-form generation is untested. (iv) Group-conditional control assumes the declared buckets capture the shift axis; unmodeled shift dimensions can still break it.

7. Conclusion

The decision to retrieve in adaptive RAG is a risk-controlled choice, and treating it as one pays off. We reframed the retrieve/skip gate as risk-controlled selective prediction under asymmetric cost, and conformalized the gate decision rather than the generated answer with Conformal Risk Control. This yielded a finite-sample, distribution-free bound on the hallucination-causing harmful-skip rate that existing adaptive-RAG gates lack. Across two model families and two datasets, the realized rate tracked the target dial from 0.02 to 0.20, spending the budget rather than hiding behind needless retrieval. We showed that the prevailing fixed-threshold practice overshoots the hallucination budget by 34–58% under distribution shift, whereas conformal recalibration restores control automatically. The skip rate auto-scaled with model competence, so that a more capable model safely declined more retrieval under the same guarantee. We further delineated when group-conditional calibration is and is not required, finding marginal control sufficient with a conditionally calibrated score and reserving Mondrian for known label shift, and validated the gate end-to-end with a live retriever as a guaranteed cost/accuracy dial. An operator who can state an acceptable hallucination-from-skipping rate now obtains the cheapest gate that respects it, with a finite-sample guarantee and automatic adaptation to the deployment distribution. This reframing of the gate as a conformalized, asymmetric-cost selective-prediction decision offers knowledge-based RAG systems a principled safety knob in place of a hand-tuned threshold.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The experiments use the publicly available PopQA (Mallen et al., 2023) and TriviaQA (Joshi et al., 2017) benchmarks. The code and derived experimental artifacts that support the findings of this study are available from the author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) to improve the language and readability of portions of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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